

2025-07-15

# BUS-STOP COOLING INTERVENTION REPORT

**AOI:** 6,713 tiles at 80 m · 40 stops · top 3 diagnosed in detail

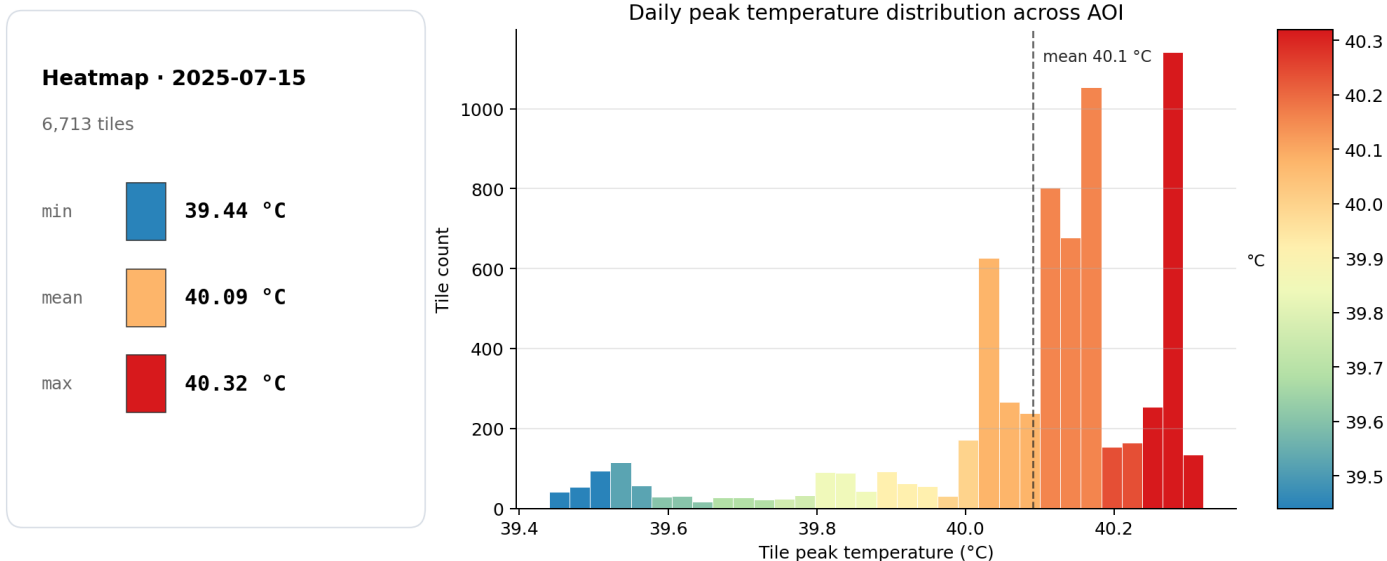
**AOI peak temperature range:** 39.4 – 40.3 °C (mean 40.1 °C)



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## AOI PEAK-TEMPERATURE DISTRIBUTION

Per-tile daily peak across the AOI. The dashed line marks the AOI mean; the histogram is colored on the same spectral ramp as every map and chart in this report.



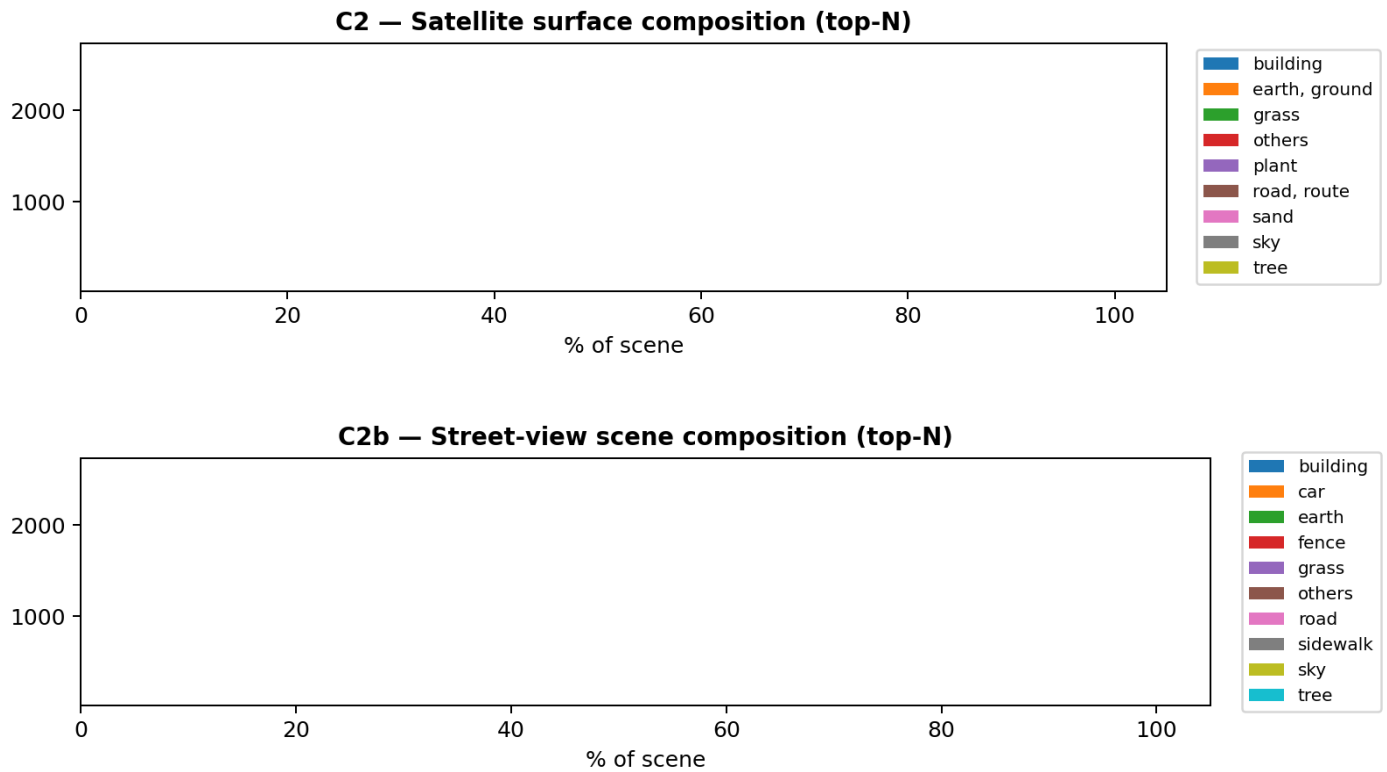
## TOP 3 HOTTEST STOPS — AT A GLANCE

Daily peak temperature, surface diagnosis (% impervious vs. vegetation), sky exposure, and peak heat index for each of the top-N stops.

| Rank | Stop ID | Name                     | Peak °C | Imperv % | Veg % | Sky % | HI °C | RH % | Wet-bulb ° |
|------|---------|--------------------------|---------|----------|-------|-------|-------|------|------------|
| #1   | 2609    | 7th Av & Tamarisk St     | 40.3    | 78       | 13    | 44    | 47.9  | 38   | 19.2       |
| #2   | 156     | Broadway Rd & Central Av | 40.3    | 22       | 4     | 7     | 46.0  | 35   | 22.7       |
| #3   | 148     | Broadway Rd & 4th St     | 40.3    | 65       | 4     | 45    | 46.0  | 35   | 22.7       |

## SURFACE COMPOSITION ACROSS TOP-N (C2 / C2B)

Stacked bars show the per-stop class breakdown for satellite (overhead) and street view (front-of-stop). Together they pin down whether the heat driver is impervious surface, low canopy, or open sky.



## #1 · 2609 — 7TH AV & TAMARISK ST

Peak temp: 40.3 °C · Lat/Lon: 33.4043, -112.0819

Surface: impervious 77.8% · vegetation 13.2%

Street view: sky 43.5% · building 5.8%

Peak heat index: 47.9 °C · RH at peak: 38.1% · Wet-bulb: 19.2 °C

### SATELLITE — ORIGINAL & SEGMENTATION

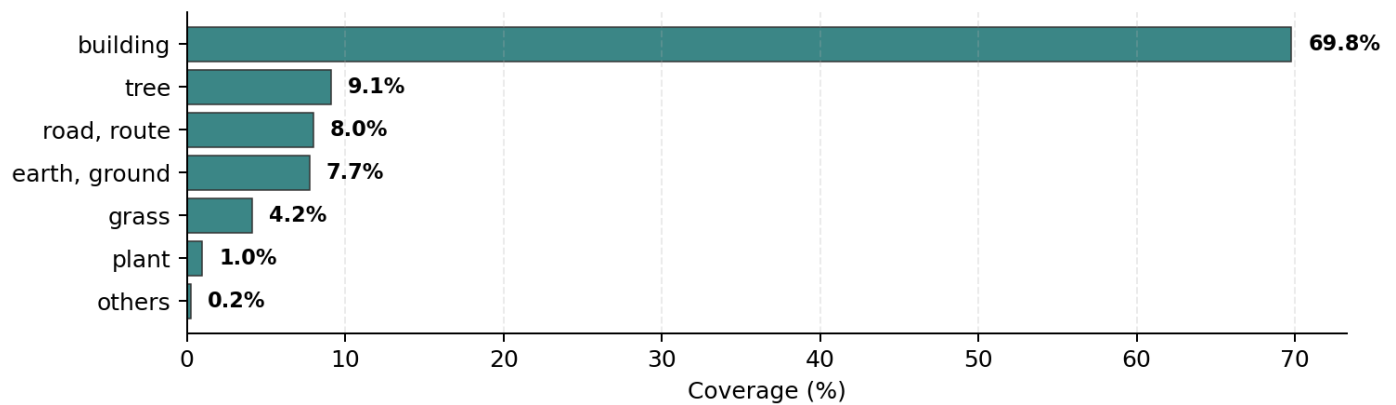


original satellite tile



segmentation overlay

#### Satellite class breakdown — 2609



## #1 · 2609 — 7TH AV & TAMARISK ST (STREET VIEW)

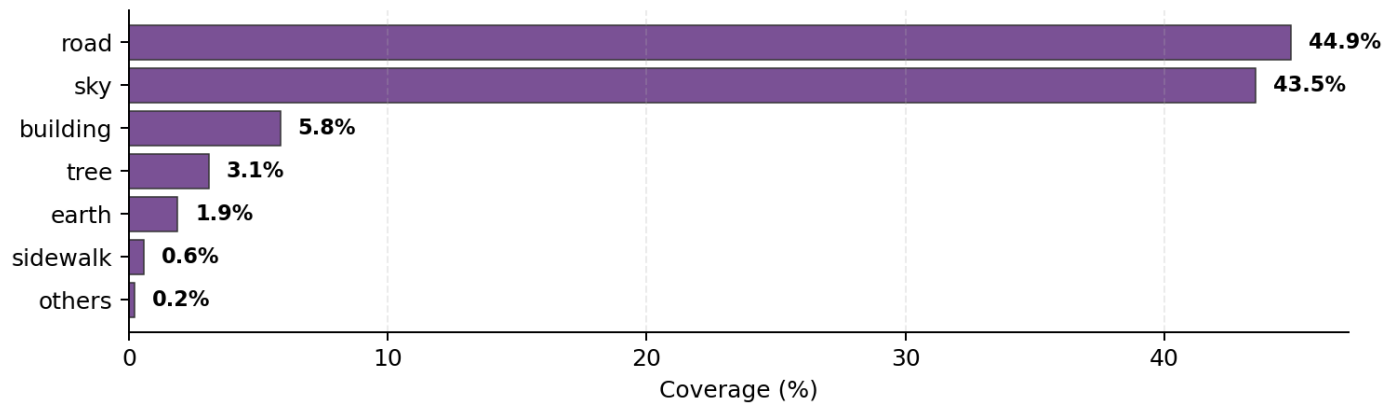


original street-view (front)



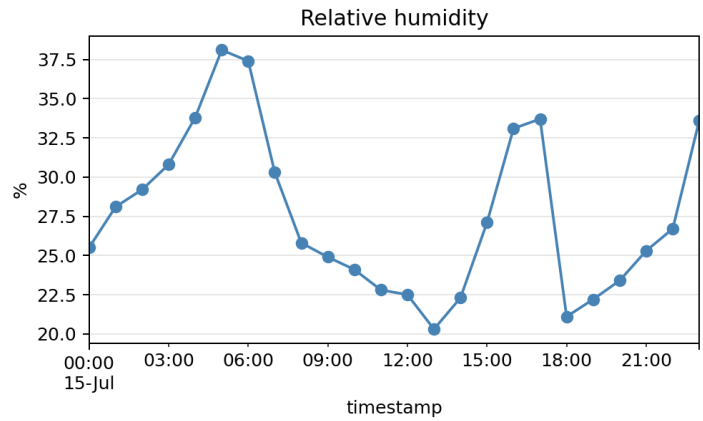
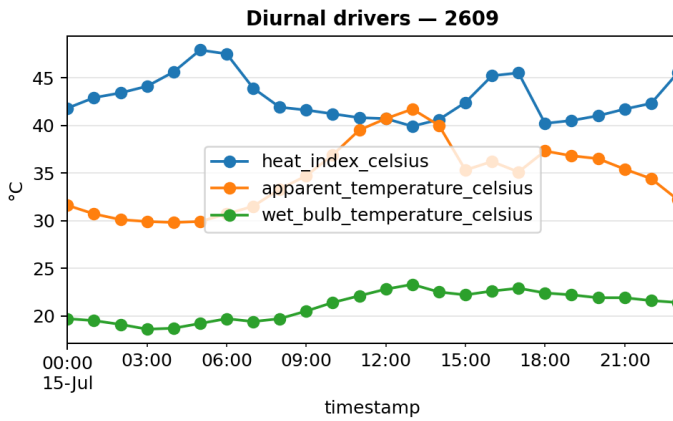
pixel-wise segmentation

### Street-view class breakdown — 2609



## #1 · 2609 — 7TH AV & TAMARISK ST (ENV-PARAMS)

Diurnal heat-index, apparent temperature, wet-bulb, and RH curves across the day.



## #2 · 156 — BROADWAY RD & CENTRAL AV

Peak temp: 40.3 °C · Lat/Lon: 33.4065, -112.0726

Surface: impervious 22.1% · vegetation 3.8%

Street view: sky 7.2% · building 3.4%

Peak heat index: 46.0 °C · RH at peak: 34.6% · Wet-bulb: 22.7 °C

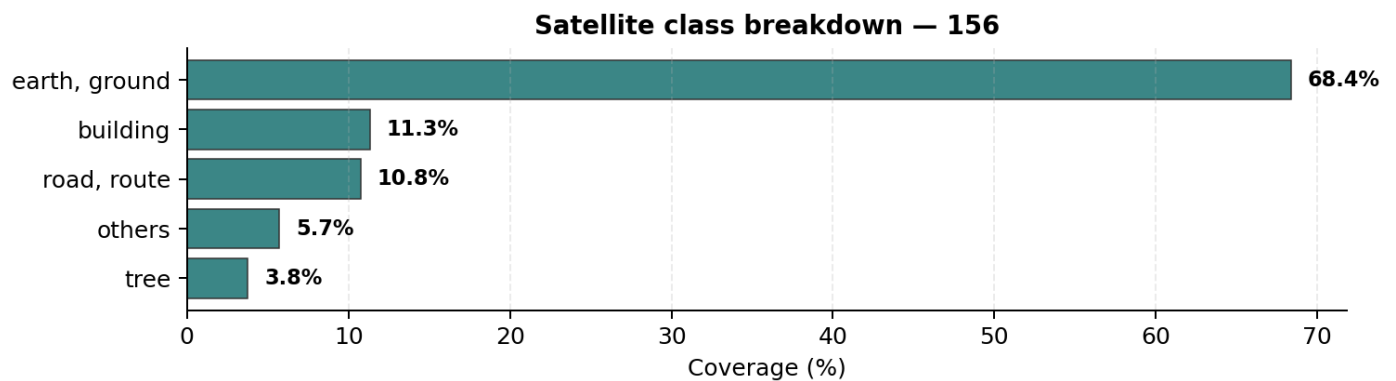
### SATELLITE — ORIGINAL & SEGMENTATION



original satellite tile



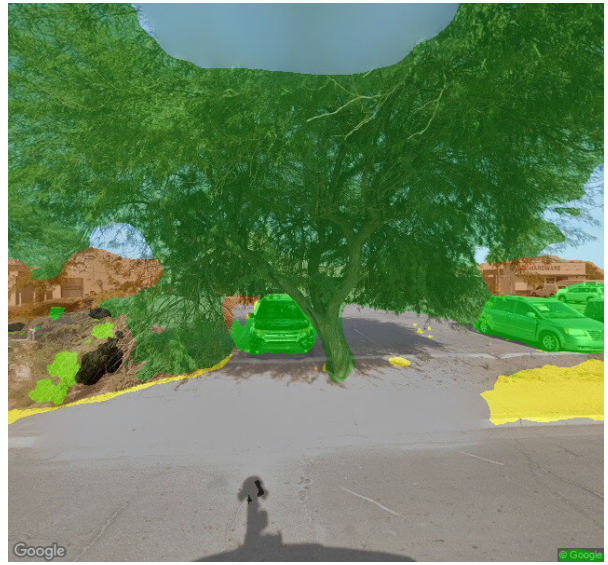
segmentation overlay



## #2 · 156 — BROADWAY RD & CENTRAL AV (STREET VIEW)

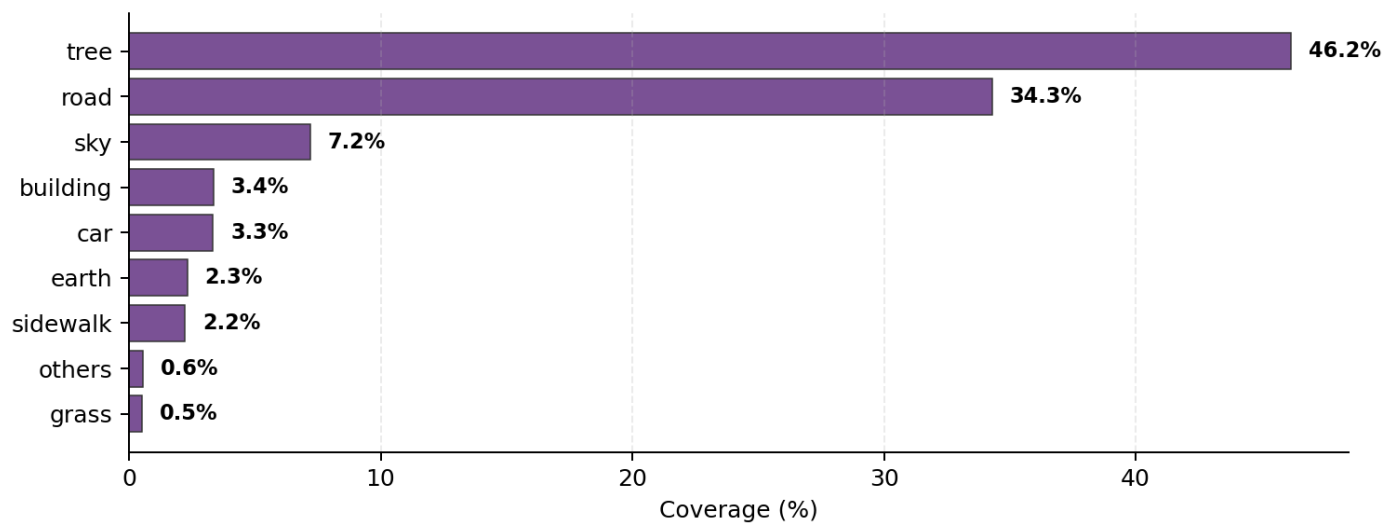


original street-view (front)



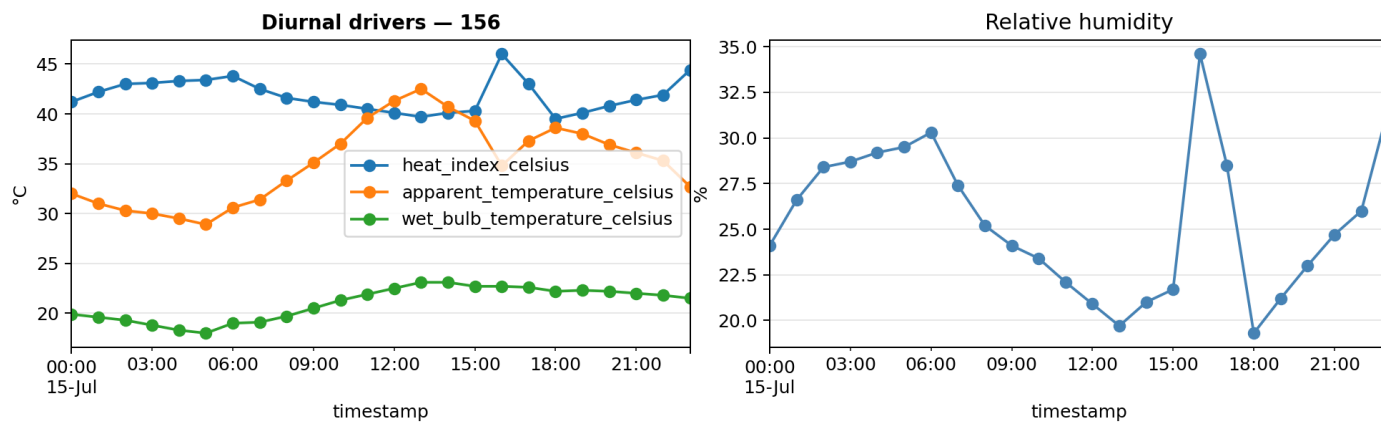
pixel-wise segmentation

### Street-view class breakdown — 156



## #2 · 156 — BROADWAY RD & CENTRAL AV (ENV-PARAMS)

Diurnal heat-index, apparent temperature, wet-bulb, and RH curves across the day.



### #3 · 148 — BROADWAY RD & 4TH ST

Peak temp: 40.3 °C · Lat/Lon: 33.4066, -112.0686

Surface: impervious 65.4% · vegetation 3.7%

Street view: sky 44.9% · building 8.5%

Peak heat index: 46.0 °C · RH at peak: 34.6% · Wet-bulb: 22.7 °C

#### SATELLITE — ORIGINAL & SEGMENTATION

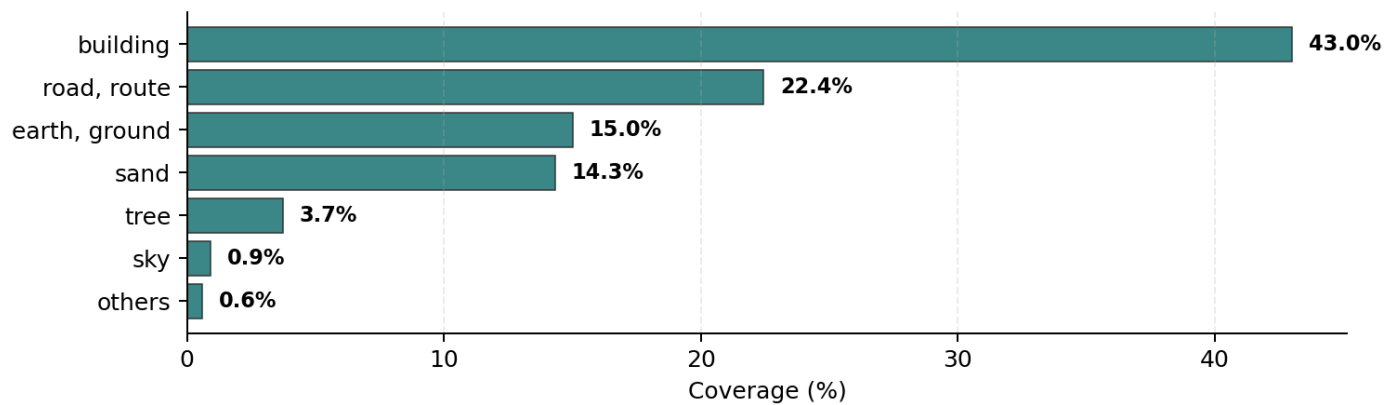


original satellite tile

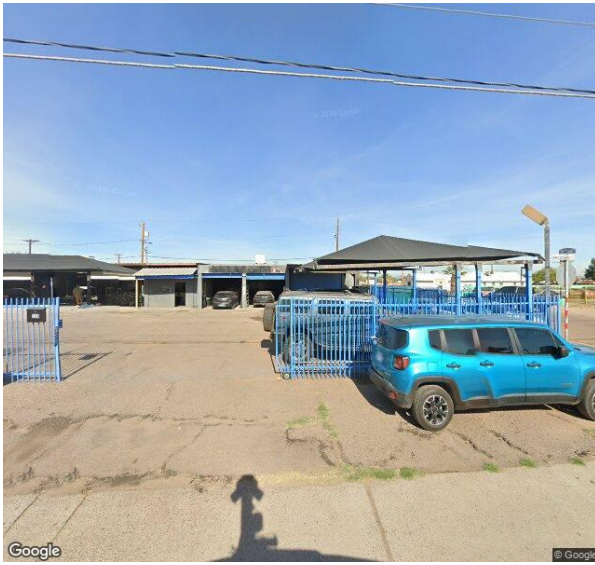


segmentation overlay

#### Satellite class breakdown — 148



### #3 · 148 — BROADWAY RD & 4TH ST (STREET VIEW)

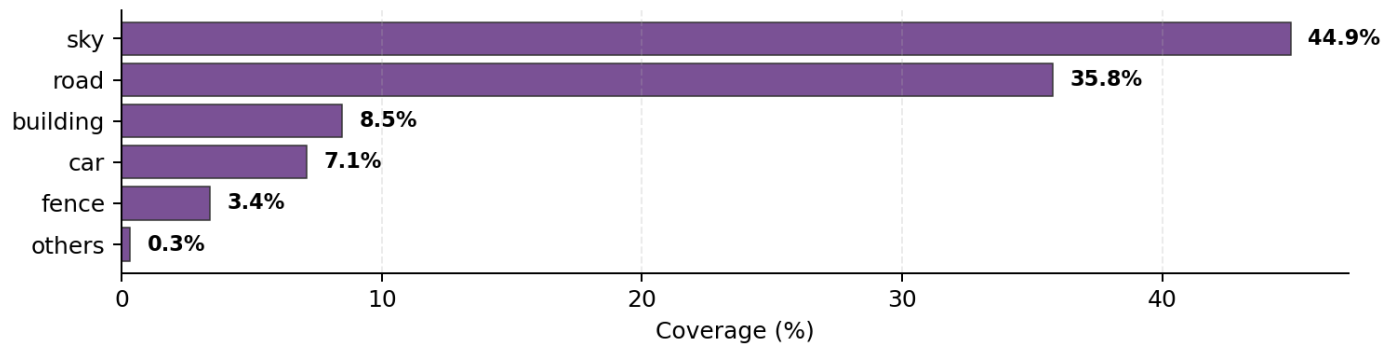


original street-view (front)



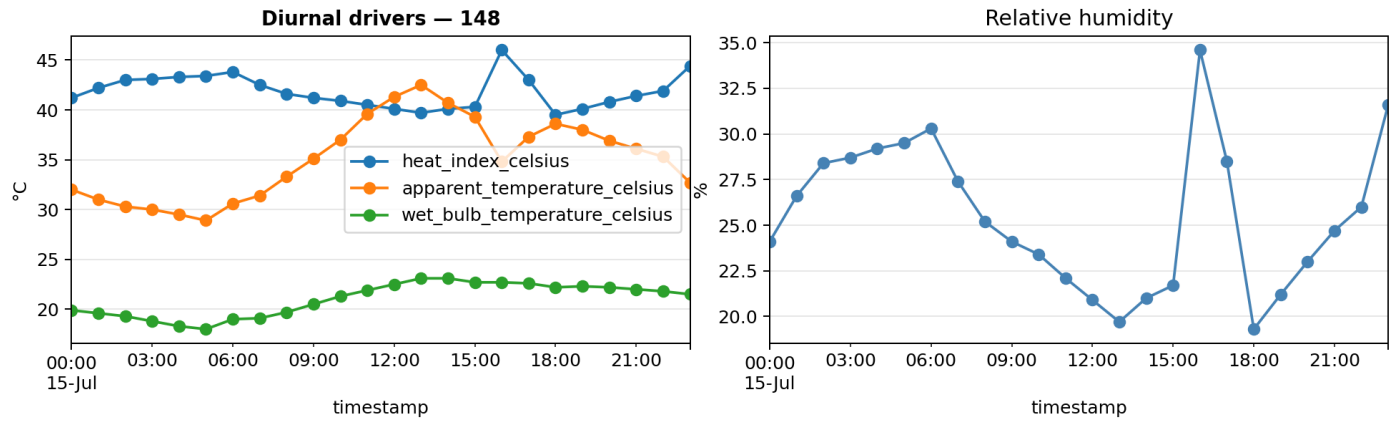
pixel-wise segmentation

#### Street-view class breakdown — 148



### #3 · 148 — BROADWAY RD & 4TH ST (ENV-PARAMS)

Diurnal heat-index, apparent temperature, wet-bulb, and RH curves across the day.



## PRIORITIZED ACTION LIST

Threshold-triggered intervention recommendations from Step 9. The primary action is the highest-leverage measure given each stop's measurements; the secondary is the next most relevant.

| Rank | Stop ID | Name                     | Peak °C | Primary action                                                       | Secondary action                                                     |
|------|---------|--------------------------|---------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| 1    | 2609    | 7th Av & Tamarisk St     | 40.3    | Install misting station (evaporative cooling — effective in dry air) | Apply reflective / cool-surface paving around shelter footprint      |
| 2    | 156     | Broadway Rd & Central Av | 40.3    | Install misting station (evaporative cooling — effective in dry air) | Plant street trees along pedestrian approach (3–5 yr cooling payoff) |
| 3    | 148     | Broadway Rd & 4th St     | 40.3    | Install misting station (evaporative cooling — effective in dry air) | Plant street trees along pedestrian approach (3–5 yr cooling payoff) |

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